

RS & WS Discrepancies?

What is going on?

Edward Wardell

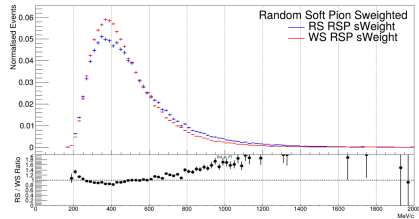
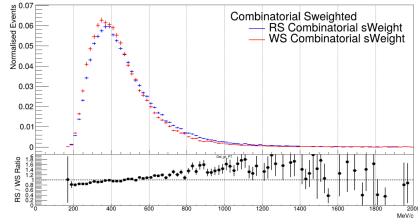
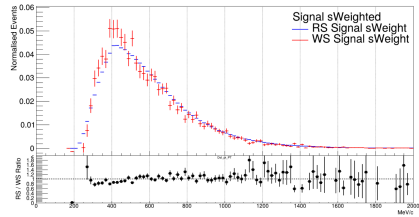
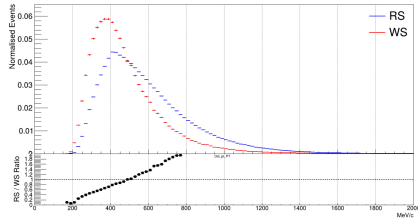
LHCb Collaboration

December 15, 2024

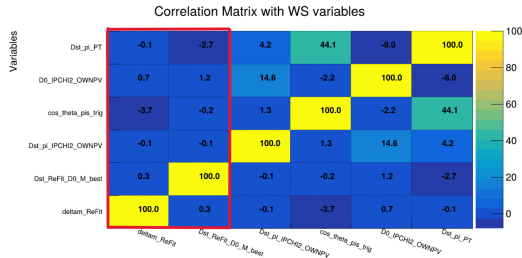
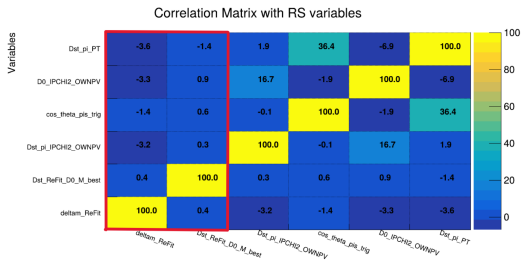
Data Considered

- ▶ Data from: Run 2, 2015 MagDown, (287798 WS events), (4446880 RS events).
- ▶ Standard analysis cuts taken.
- ▶ sWeighting was obtained using a 2D invariant mass plot, using $\Delta m = m(D^*) - m(D^0)$, using both reconstructed $m(D^*)$ & $m(D^0)$. VS the reconstructed $m(D^0)$.
- ▶ sWeighting requires the discriminant and control variables to be independent.

sWeighting **Dst_pi_PT**: (Soft Pion Transverse Momenta)



Pearson's Correlation Coefficient Plots with Invariant Masses

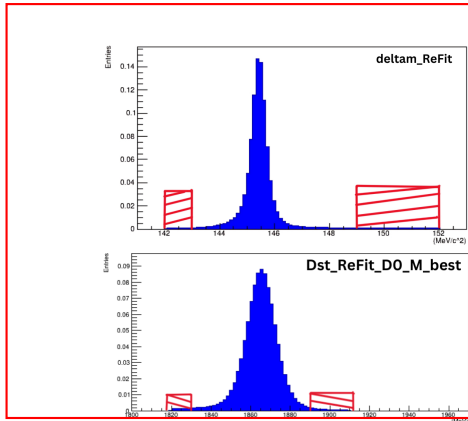


Combinatorial Region from Mass Windows:

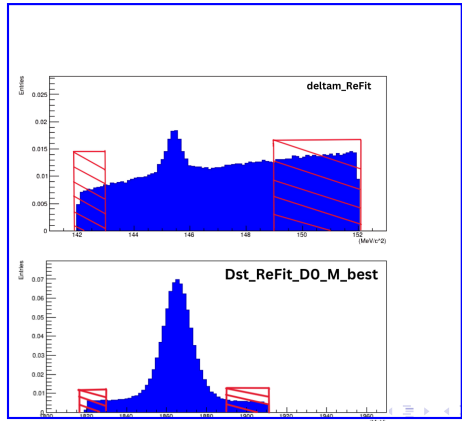
$(143 \geq \text{deltam_ReFit} \text{ or } 149 \leq \text{deltam_ReFit}) \text{ \& }$

$(1830 \geq \text{Dst_ReFit_D0_M_best} \text{ or } 1890 \leq \text{Dst_ReFit_D0_M_best})$

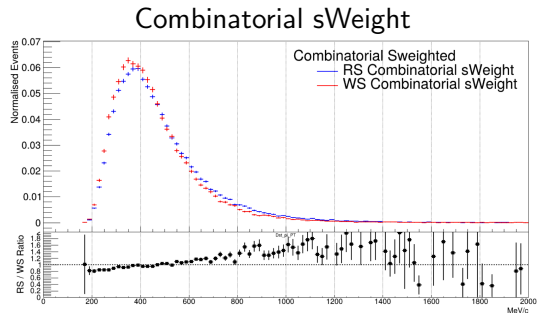
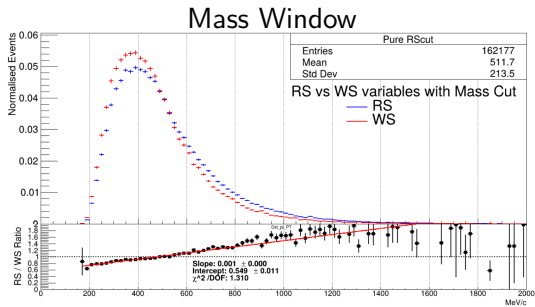
RS



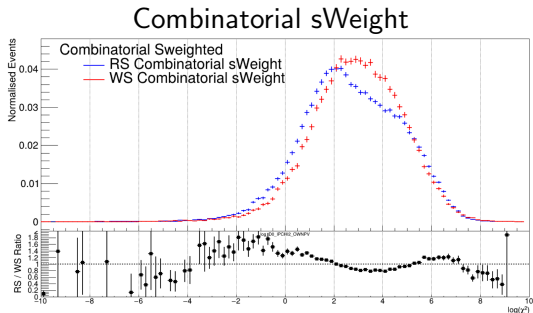
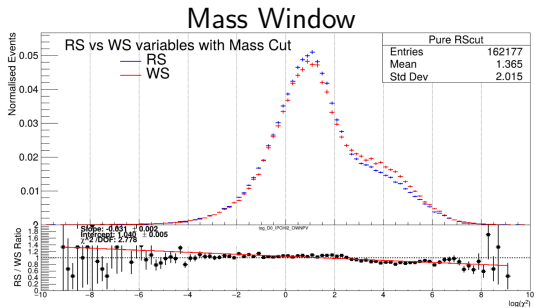
WS



Dst_pi_PT Combinatorial Region from Mass Windows

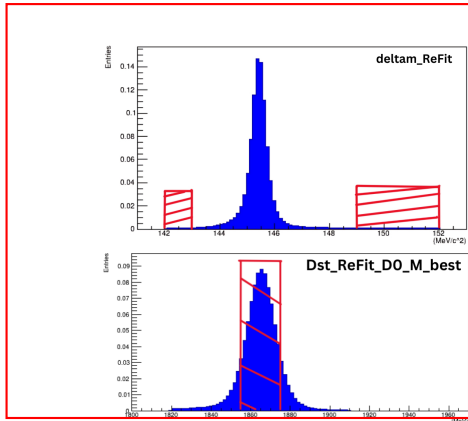


log(D0_IPCHI2_OWNPV): (χ^2 of primary vertex of D0)

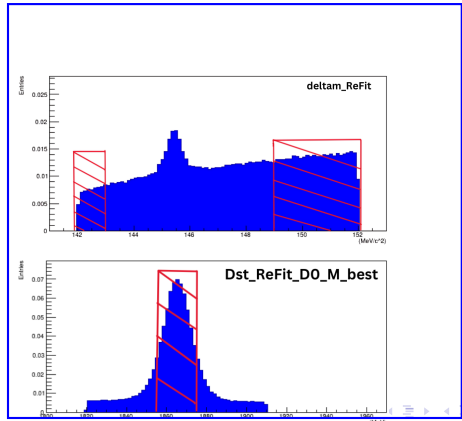


Approximate RSP Region from Mass Windows:
 $(143 \geq \text{deltam_ReFit} \text{ or } 149 \leq \text{deltam_ReFit}) \ \& \$
 $(1855 \leq \text{Dst_ReFit_D0_M_best} \leq 1875)$

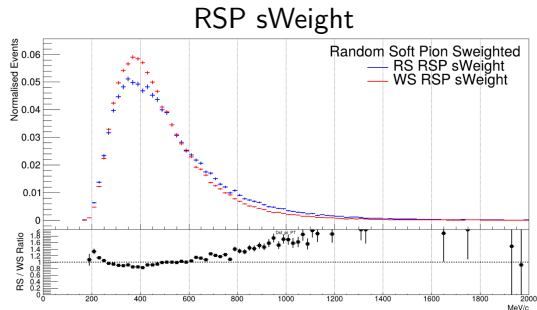
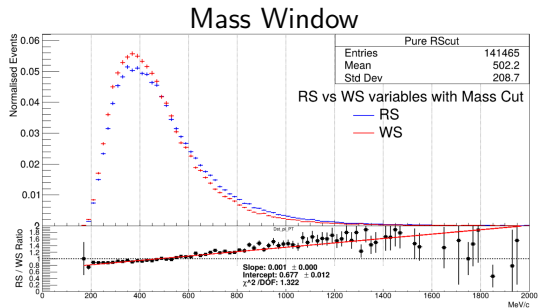
RS



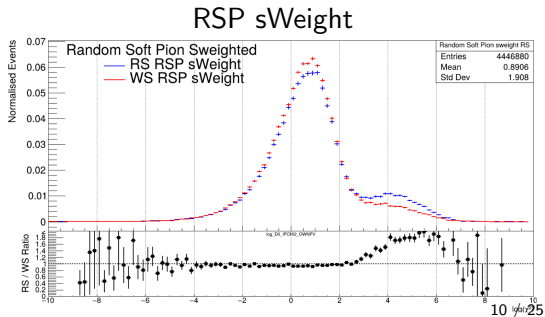
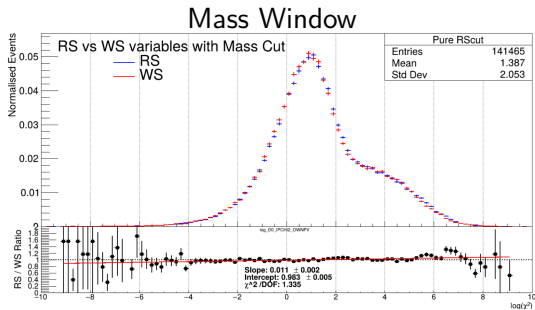
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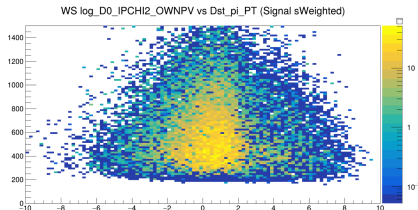
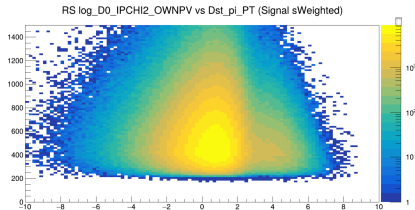
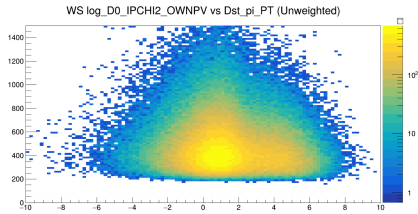
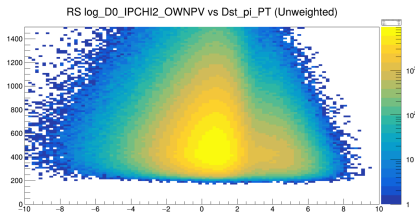
Dst_pi_PT Combinatorial Region from Mass Windows



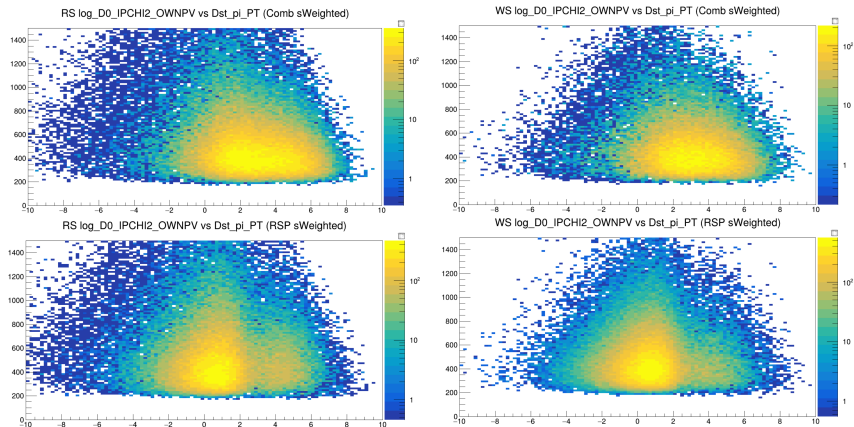
log(D0_IPCHI2_OWNPV): (χ^2 of primary vertex of D0) **Combinatorial Region from Mass Windows**



Dst_pi_PT vs log(D0_IPCHI2_OWNPV)

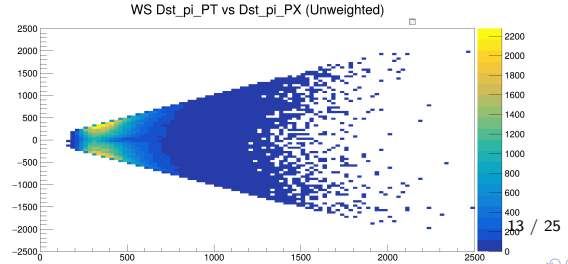
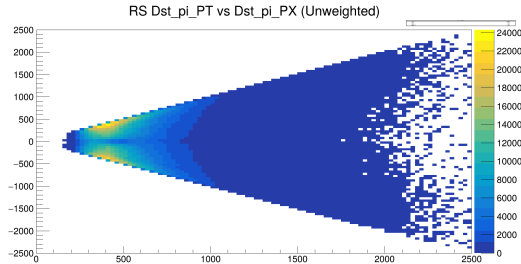


Dst_pi_PT vs log(D0_IPCHI2_OWNPV)

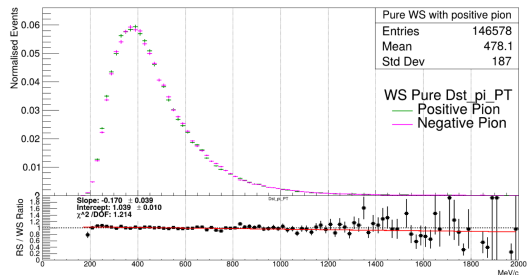
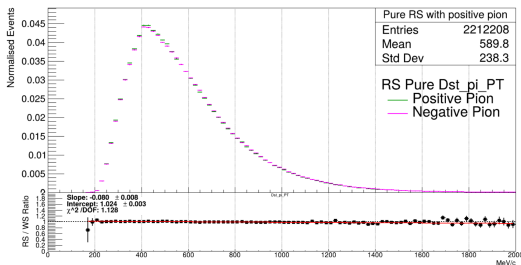


Detector Differences?

- ▶ Asymmetries in the detector are mainly due to the following reasons:
- ▶ Initial state (pp collisions) at the LHC is not CP symmetric.
- ▶ LHCb detector is not perfectly left-right symmetric
- ▶ Pattern recognition, track reconstruction, and selections are intrinsically asymmetric since the detector is made of matter.



Differences between π_S^+ & π_S^- . No discrepancies found here, or indeed in deltam_ReFit regions $[145.25, 145.80]$, $\text{deltam_ReFit} \geq 148$. or $\text{deltam_ReFit} \leq 143$.



Conclusion

- ▶ It seems that there is an intrinsic difference between the RS and WS data.
- ▶ Questions?
- ▶ Thank your for listening

Extra Slides

How the Variables are Normalised

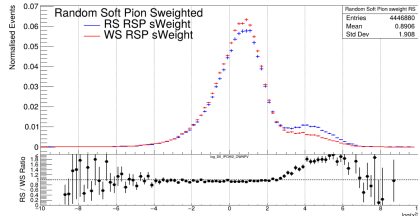
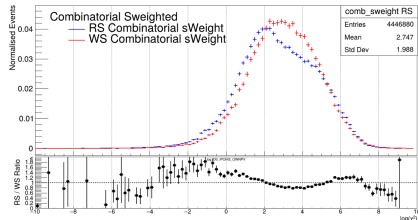
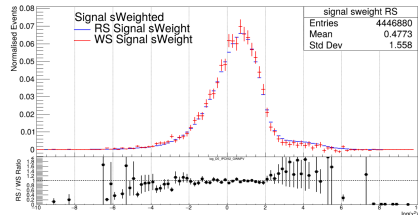
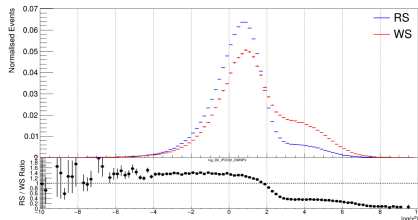
```
def normalise_histogram(h: TH1D) -> None:
    """
    Normalizes a histogram.

    Args:
        h (TH1D): The histogram to normalise.
    """
    integral: float = h.Integral()
    if integral != 0:
        h.Scale(1.0 / integral)
    else:
        print(f"Histogram {h.GetName()} is empty and cannot be normalised.")
```

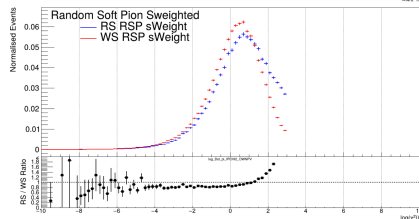
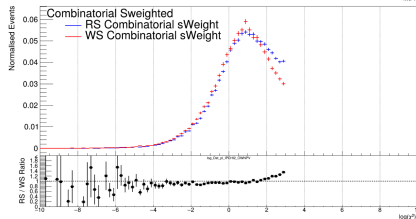
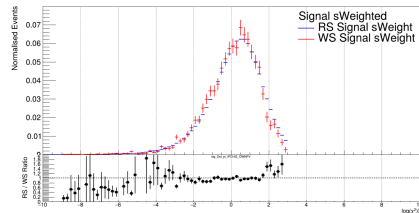
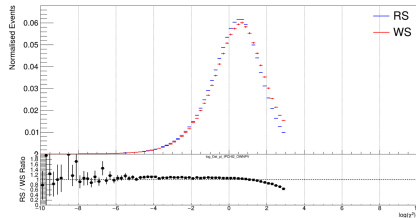
Pearson's Correlation Coefficient

- ▶ Calculated via: $corr(x, y) = \frac{cov(x, y)}{\sigma_x \sigma_y}$
- ▶ $cov(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)$
- ▶ $\sigma_x = \sqrt{\frac{1}{\sum_{i=1}^N (x_i - \mu_x)^2}}$
- ▶ Doesn't take into account factorisation
- ▶ Factorisation means independence (which is more than just not-linearly-correlated)

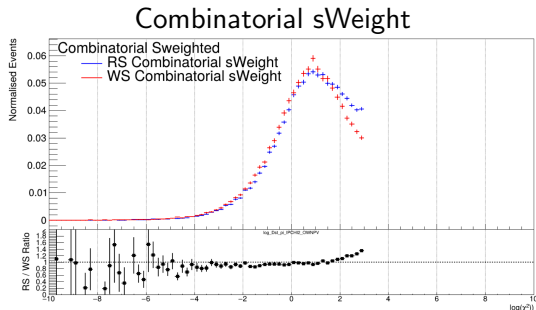
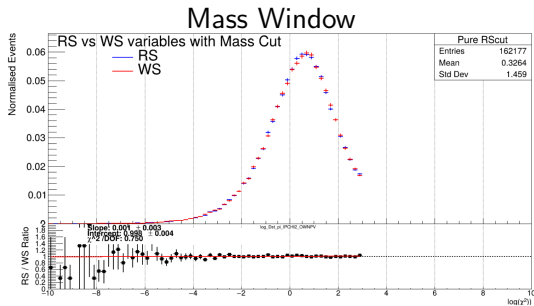
sWeighting $\log(D0_IPCHI2_OWNPV)$: (χ^2 of primary vertex of $D0$)



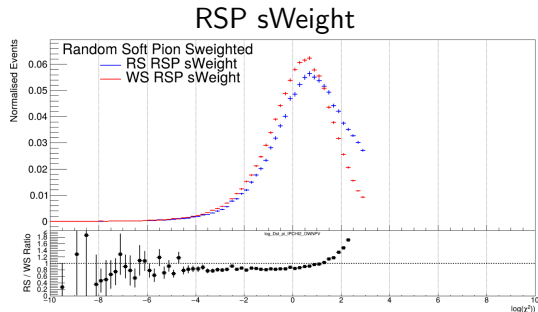
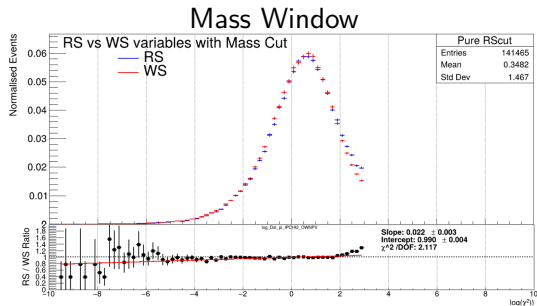
sWeighting $\log(\text{Dst_pi_IPCHI2_OWNPV})$: (χ^2 of primary vertex of π_s)



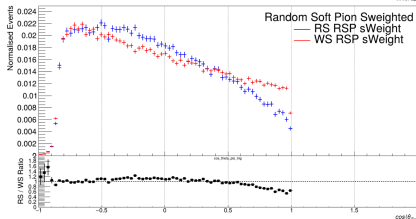
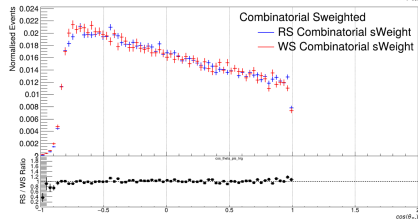
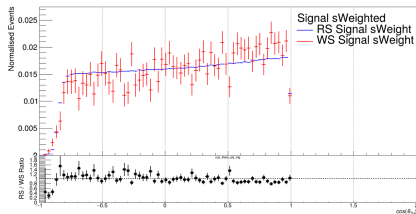
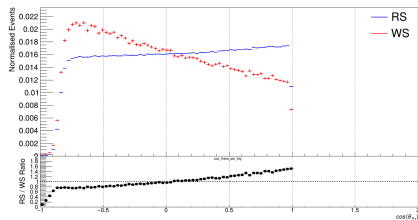
$\log(\text{Dst_pi_IPCHI2_OWNPV})$ Combinatorial Region from Mass Windows



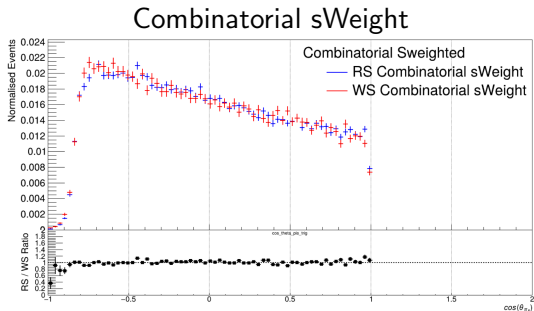
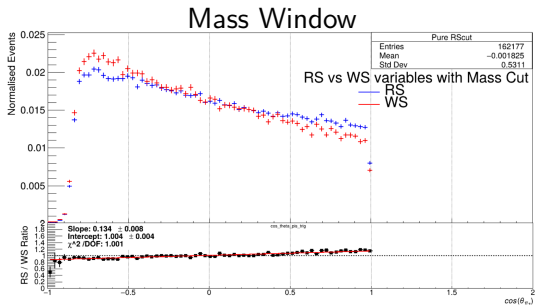
$\log(\text{Dst_pi_IPCHI2_OWNPV})$ RSP Region from Mass Windows



sWeighting **cos_theta_pis_trig**: (*Cosine of the Helicity Angle of π_s*)



$\cos_{\text{theta_pis_trig}}$ Combinatorial Region from Mass Windows



$\cos\theta_{\pi^0\pi^0\text{trig}}$ RSP Region from Mass Windows

